



CS 260 – Foundations of Data Science

Lecture 7 – Evaluation Metrics & Probability

09/22/2026

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Announcements

HW3 due Tuesday night - tonight

Midterm 1 – October 6th

Office Hours:

Tuesdays: 4pm-4:30pm

Thursdays 2-3:30pm, Park 200D



Confusion Matrix

True Positive: How many times was $\hat{y} = 1$ and $\hat{y} = y$

True Negative: How many times was $\hat{y} = 0$ and $\hat{y} = y$

False Positive: How many times was $\hat{y} = 1$ and $\hat{y} \neq y$

False Negative: How many times was $\hat{y} = 0$ and $\hat{y} \neq y$

Confusion Matrix

Probability of an event **e**

$$P(e) = \frac{\textit{count}(e)}{\textit{count}(\textit{all events})}$$

If we put all events in a bag, shake it up, and choose one at random, how likely are we to get **e**

This is called **sampling**

Intro to Probability



- Suppose we flip a fair coin
- What is the probability of heads, $P(e = H)$?



Intro to Probability



- Suppose we flip a fair coin
- What is the probability of heads, $P(e = H)$?
- We have "all" of two possibilities, $e \in \{H, T\}$.
- $$P(e = H) = \frac{\text{count}(H)}{\text{count}(H) + \text{count}(T)}$$



Intro to Probability



- Suppose we have a fair 6-sided die.
- What's the probability of getting "1"?



Intro to Probability



- Suppose we have a fair 6-sided die.
- What's the probability of getting "1"?

$$\frac{\textit{count}(s)}{\textit{count}(1) + \textit{count}(2) + \textit{count}(3) + \cdots + \textit{count}(6)} = \frac{1}{1 + 1 + 1 + 1 + 1 + 1} = \frac{1}{6}$$



Intro to Probability



- What about a die with only three numbers $\{1, 2, 3\}$, each of which appears twice?
- What's the probability of getting "1"?



Intro to Probability



- What about a die with only three numbers $\{1, 2, 3\}$, each of which appears twice?
- What's the probability of getting "1"?

$$P(e = 1) = \frac{\text{count}(1)}{\text{count}(1) + \text{count}(2) + \text{count}(3)} = \frac{2}{2 + 2 + 2} = \frac{1}{3}.$$



Intro to Probability



- The set of all probabilities for an event e is called a **probability distribution**
- Each coin toss is an independent event (Bernoulli trial).



Intro to Probability



- Which is greater, $P(HHHHHH)$ or $P(HHTHHH)$?



Intro to Probability



- Which is greater, $P(HHHHHH)$ or $P(HHTHHH)$?
- Since the events are independent, they're equal



Intro to Probability

Probability Axioms

1. Probabilities of events must be no less than 0. $P(e) \geq 0$ for all e .
2. The sum of all probabilities in a distribution must sum to 1. That is,
 $P(e_1) + P(e_2) + \dots + P(e_n) = 1$. Or, more succinctly,

$$\sum_{e \in E} P(e) = 1.$$



Joint Probability

The probability that two independent events e_1 and e_2 both occur is the product of $P(e_1) * P(e_2)$

$$P(e_1 \wedge e_2) = P(e_1 \cap e_2) = P(e_1)P(e_2) \text{ when } e_1 \cap e_2 = \emptyset$$

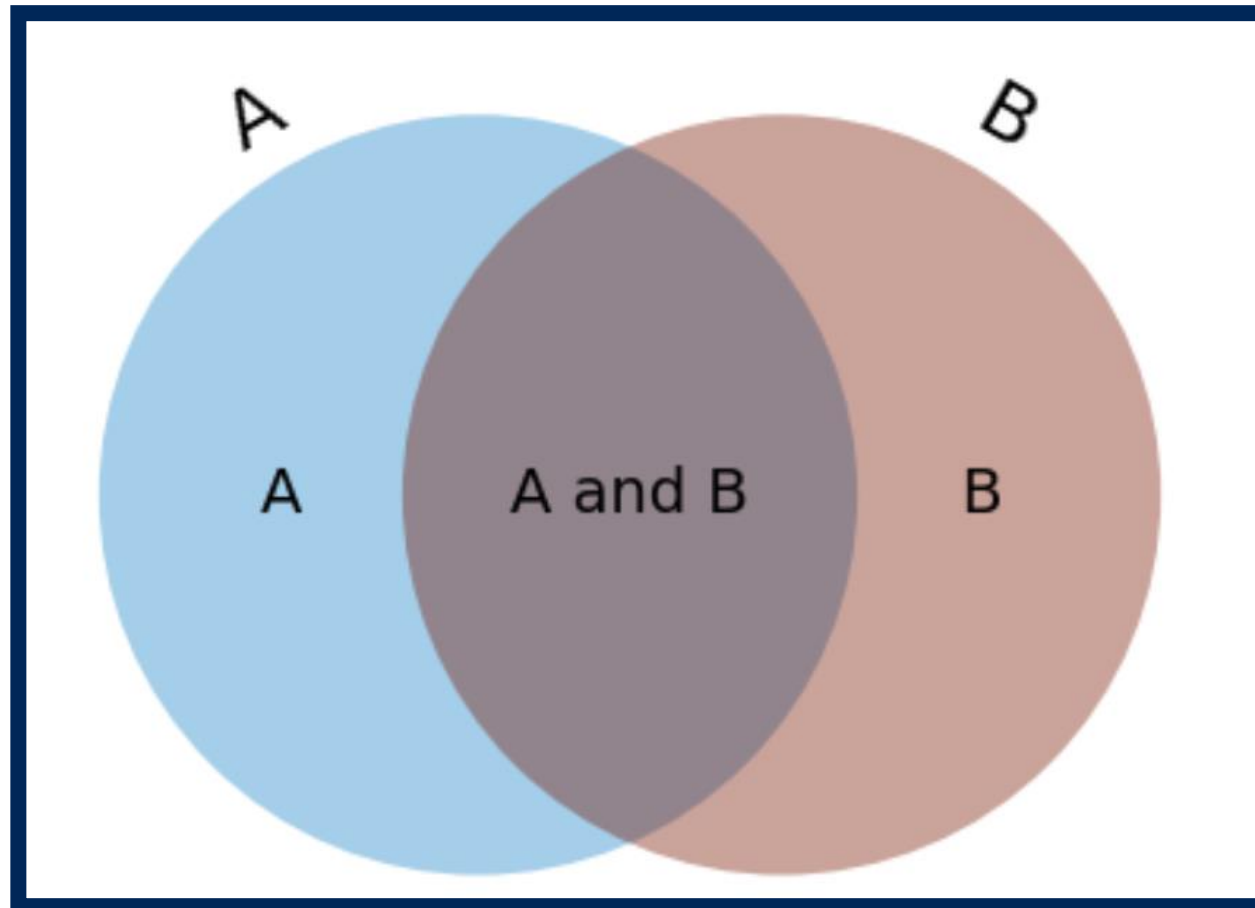
$P(e_1)$ - fraction of probability space occupied by event e_1

- So $P(e_1 \wedge e_2)$ is the fraction of e_1 probability space where e_2 where also occurs
- If $P(e_1) = \frac{1}{2}$ and $P(e_2) = \frac{1}{3}$, then $P(e_1, e_2)$ is a third of a half of the probability space



Intro to Probability

Joint Probability



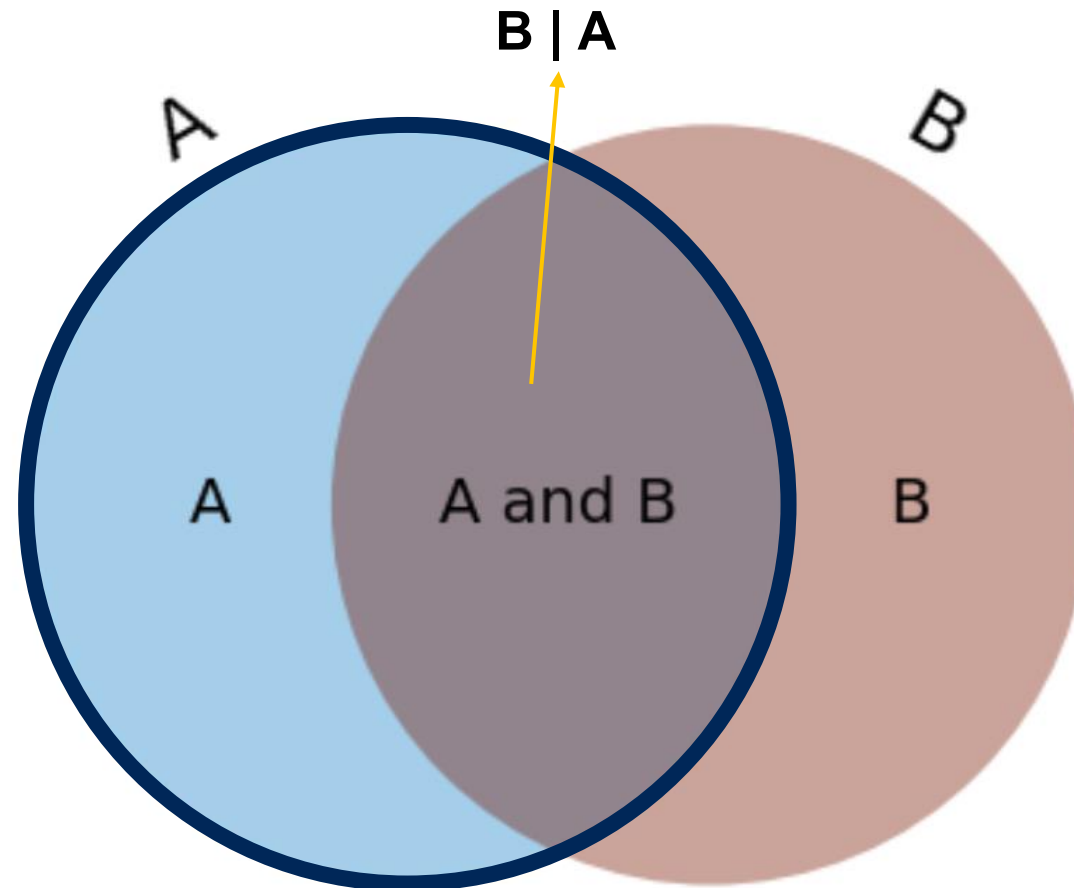
Conditional Probability

The probability that one event occurs given that we take the other for granted

The probability of e_2 given e_1 is $P(e_2 | e_1)$

Intro to Probability

Conditional Probability



Marginal Probability Distributions

Given a discrete joint probability distribution function $P(X, Y)$, how would we find $P(X)$?

Marginalize out the Y – sum over all $y \in Y$

$$\text{Discrete case: } p(x) = \sum_{\{y \in Y\}} P(x, y)$$

$$\text{Continuous Case } p(x) = \int p(x, y) dy$$



Intro to Probability

Marginal Probability Distributions

Example: $P(u) = P(u, \text{rain}) + P(u, \overline{\text{rain}})$

